

A *left rotation* operation on a circular array shifts each of the array's elements **1** unit to the left. The elements that fall off the left end reappear at the right end. Given an integer *d*, rotate the array that many steps to the left and return the result.

Example

d = 2
arr = [1, 2, 3, 4, 5]

After 2 rotations, *arr'* = [3, 4, 5, 1, 2].

Function Description

Complete the *rotateLeft* function with the following parameters:

- *int d*: the amount to rotate by
- *int arr[n]*: the array to rotate

Returns

- *int[n]*: the rotated array

Input Format

The first line contains two space-separated integers that denote *n*, the number of integers, and *d*, the number of left rotations to perform.
The second line contains *n* space-separated integers that describe *arr*[].

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq d \leq n$
- $1 \leq a[i] \leq 10^6$

Sample Input

STDIN	Function
-----	-----
5 4	n = 5 d = 4
1 2 3 4 5	arr = [1, 2, 3, 4, 5]

Sample Output

5 1 2 3 4

Explanation

To perform $d = 4$ left rotations, the array undergoes the following sequence of changes:

$$[1, 2, 3, 4, 5] \rightarrow [2, 3, 4, 5, 1] \rightarrow [3, 4, 5, 1, 2] \rightarrow [4, 5, 1, 2, 3] \rightarrow [5, 1, 2, 3, 4]$$